

# A Test Method for Self-driving Vehicle Based on Mixed Reality

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**Abstract**—This paper proposes a mixed reality-based self-driving vehicle testing method, aiming at self-driving vehicle safety testing in a complex environment. This method firstly maps the virtual scene in the simulation environment to the real test environment. Secondly, the performance indexes such as intelligent perception and the self-driving test vehicle's behavior decision are quickly collected through sensor and data fusion. A closed-loop test system is then constructed based on the mixed reality test scene, control center, and test vehicle. Finally, through the case verification, it shows that this method can meet self-driving vehicles' test requirements in various scenarios.

**Keywords**- mixed reality; self-driving; test method; test scene; background vehicle

## I. INTRODUCTION

The self-driving vehicle is an important carrier for applying new technologies such as artificial intelligence, big data, and high-performance computing. It significantly impacts reducing traffic congestion, improving traffic safety and travel efficiency, and constructing intelligent city traffic planning. At present, it has been popularized and applied in trunk logistics, closed parks, pilotless buses, pilotless sanitation vehicles, and other scenarios. However, with the continuous exposure of self-driving accidents, people question self-driving safety[1]. Testing and evaluating self-driving vehicles' safety in complex traffic scenarios has become a problem that many researchers have to consider. At present, vehicle testing mainly includes laboratory testing and road testing. Because road testing can directly reflect vehicles' performance in actual road conditions, it is an indispensable link for self-driving vehicles before they go on the road[2]. According to the different test areas, road testing can be divided into closed or open road tests. A closed site is a completely closed environment that simulates the actual driving scene in a specific area. Closed site testing has high safety, but the test scene is limited and can not completely cover the natural environment faced by a self-driving vehicle[3]. Open road testing is a test method to verify vehicles' self-driving ability in the section area where the government allows the test to be carried out; its disadvantage

is that it easy to be affected by the actual traffic conditions, weather conditions, and many accidental events, and it has considerable risk[4,5].

A variety of mature testing methods have been developed in the traditional automotive and ADAS(Advanced Driver Assistance System) fields, and It mainly includes real vehicle tests, virtual simulation tests, hardware-in-the-loop tests, and the whole vehicle-in-the-loop test[6-9]. Real vehicle test is required before mass production, And at least need to pass hundreds of thousands of kilometers of real driving tests in various traffic scenarios. The single virtual simulation test method has the advantages of low test cost and strong repeatability. However, the model's authenticity is relatively low, and there is a big gap between the simulation results and the actual situation. Also, it can not realize the vehicle hardware performance test and the calibration[10]. In response to the above, Hardware-in-the-loop(HIL) is widely used[11,12]. HIL method combines physical controllers with virtual objects. Virtual objects are modeled according to vehicle dynamics and connected with the physical controller to form closed-loop control; due to the introduction of resistance, friction, and other vital parameters, HIL has high confidence[1]. Compared with real vehicle tests, it has the advantages of low cost, short development period, and easy reproduction. Vehicle-in-the-Loop(VIL) method combines actual vehicles with virtual objects, Constructing simulated roads, traffic scenarios, and environmental factors under laboratory conditions, And uses real vehicles for tests[13]. Some researchers further make full use of the existing methods and tools in the loop test to form an X-in-loop test method, using the model or code to replace the missing parts to achieve the software and hardware test[14].In the past, various in-loop tests were mainly aimed at vehicle body performance, Focus on vehicle dynamics and control performance testing, such as chassis control systems ABS(Antilock Brake System) and ESC(Electronic Stability Control), and testing various actuators and ECU. In the self-driving vehicle test, the decision planning module and various sensors are added, and the function of the control module is more complex than in the past, so it puts forward higher requirements for test methods.

To sum up, the self-driving test and evaluation system is still in the research stage, and there is no uniform standard to follow in the industry. This paper attempts to introduce a new self-driving test method from the perspective of the combination of virtual reality gives the composition and

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implementation of the test system, and further explains the system's use through a group of specific cases. This method combines the advantages of a closed site test and open road test and overcomes the limitation of lack of traffic scene so that the functional test of a self-driving vehicle is adequate, safe, and reliable.

## II. SELF-DRIVING TEST METHOD BASED ON MIXED REALITY

### A. Basic Framework

Because there are many artificial elements in the loop test, it is not easy to accurately reflect the natural environment's interaction. This paper puts forward a mixed virtual simulation environment and natural traffic environment self-driving test method. This method combines simulation environment, network, and physical environment, executes repeatable and quantifiable test cases in the virtual simulation environment, and converts test cases into control instructions for background vehicles by using communication and control technology. By monitoring the self-driving test vehicle's dynamic behavior (primary vehicle) and the interaction between the vehicles, a mixed reality self-driving test system is constructed, and the self-driving performance will be systematically evaluated.

Firstly, the virtual scene and vehicle modeling, scene building, and task formulation are carried out according to the main vehicle's test requirements, and then the virtual simulation environment and control instructions are mapped to the real and virtual background vehicles.

Through the coupling operation of the background vehicle and the control center, the synchronous interaction between the control instruction and the vehicle state acts together on the self-driving test vehicle.

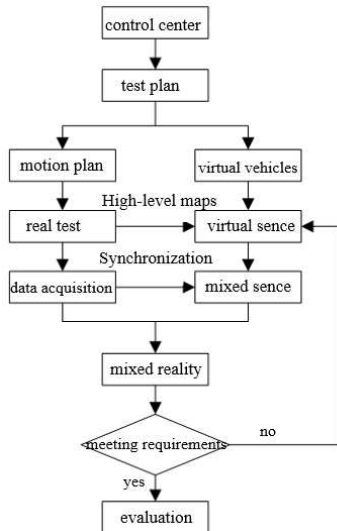


Figure 1. Flowchart of Self-driving Test Method

Finally, using the flexible set of simulation scenarios and test cases to test the main vehicle's safety performance thoroughly, the test evaluation results are output. The basic flow of the test method in this paper is shown in figure1.

### B. Mixed Reality System Construction

The self-driving test's mixed reality system is composed of

a test field, control center, and background car; figure 2 shows the system's architecture. The test field mainly provides the actual test environment and records the real-time data generated in the test process. The control center mainly monitors the test field in real-time and remotely controls the background vehicle. The mixed reality simulation environment is constructed by organically merging the real-time data with the virtual data. Finally, the mixed reality test is realized through the interaction of self-driving test vehicle, actual background vehicle, and virtual test vehicle, and the test data are analyzed and evaluated.

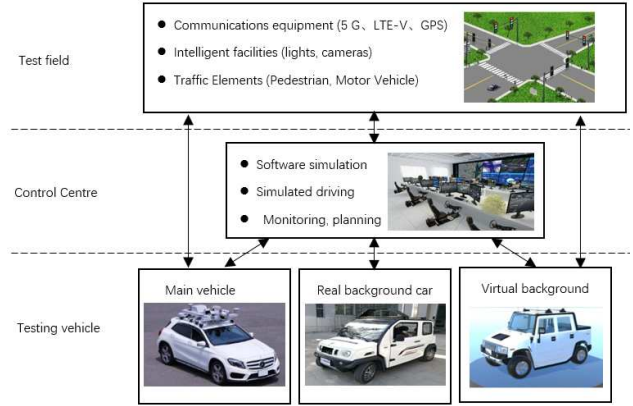


Figure 2. Schematic diagram of mixed reality system

The test field is a real test environment with a complete network connection. It forms an intelligent ecosystem[14] of "vehicle-road" and "vehicle-vehicle" instant interaction by equipped with vehicle networking communication facilities, roadside units, intelligent transportation infrastructure, and differential GPS base stations. The control center carries on the path planning, the behavior simulation, the monitoring, and the remote control to the background vehicle through the human-computer interaction to realize the self-driving test scheme under the complex traffic environment. Then the virtual simulation environment is established, the real data of the main vehicle and the background vehicle are mapped to the simulation system, and the test process under the actual scene is reproduced synchronously in the virtual environment. and build more complex or dangerous test scenarios by adding elements such as virtual vehicles in the virtual simulation environment.

### C. Self-driving Test Method

Based on the mixed reality system, the implementation of this test method is divided into four parts:

(1) Build a virtual test field corresponding to the actual test field to simulate the actual traffic environment. The virtual software provides an input interface of physical world information and an output interface of virtual world control to realize virtual simulation and physical vehicle mirror operation.

(2) Utilize props such as background vehicles, signs, and obstacles to create a typical scene (e.g., AEB test) in an actual test field and examine whether the main vehicle's behavior achieves the desired goal. And combine Real-time data of the actual test field, main vehicle, and background vehicle to reproduce the virtual test field scene.

(3) Add virtual background vehicle or sensor to generate test cases in the virtual simulation environment and convert test cases running in the simulation environment into control instructions that drive the main vehicle and actual background vehicle. Then, feedback the actual test data to the simulation environment through the network layer to realize the closed-loop feedback test of virtual reality synchronization.

Finally, using the data of the main vehicle and traffic environment collected during the test process, combined with the given test task, the self-driving vehicle's safety, comfort, and efficiency are evaluated through multi-index and multi-space-time dimensions.

### III. ADAPTIVE CRUISE CONTROL CASES

Next, the adaptive cruise control (ACC) test is taken as an example to verify the method. Self-driving vehicle ACC functions include identification, following, and speed control for the vehicle ahead. A typical ACC function test case mainly includes the following three test scenarios, as shown in figure 3.

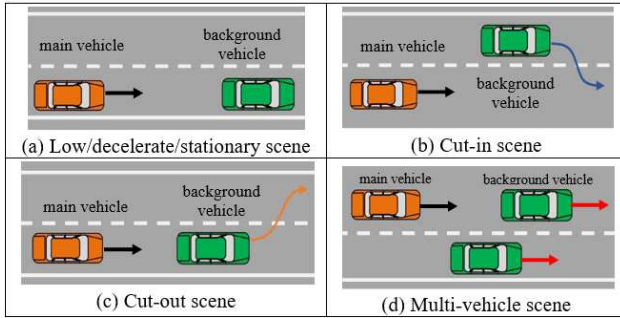


Figure 3. ACC function test scenario for self-driving vehicles

#### A. Low/decelerate/stationary scene

After the main vehicle activates the ACC, it cruises at the set speed, and the background vehicle appears in the running state of low speed, deceleration, and stationary, respectively, in the front of the main vehicle. The main vehicle should identify the background vehicle ahead. ACC controls the vehicle to slow down to avoid collision automatically.

#### B. Cut-in/out scene

After the main vehicle activates the ACC, it cruises at a set speed; the cut-in scene means that the background vehicle cuts into the main lane at a specified speed within a certain distance. The cut-out scene means that the background vehicle cut out from the main lane to the adjacent lane.

#### C. Multi-vehicle scene

After the main vehicle activates the ACC, it follows the background vehicle at a set speed and gradually approaches and exceeds another low-speed background vehicle in the adjacent lane. In this scenario, there are many background vehicles; the main vehicle can not be misidentified and continue to follow the background vehicle ahead running state unchanged.

The control function of the ACC system should meet the following basic requirements: first, the main vehicle should maintain a uniform speed or a certain workshop time distance with the front vehicle, and the vehicle acceleration and deceleration should not exceed the limit requirements should not appear excessive braking phenomenon. Secondly, the main vehicle can detect and identify the front vehicle. If there is more than one vehicle in front, the ACC system should automatically select the nearest vehicle ahead in the central lane.

### IV. TEST AND ANALYSIS

#### A. Test process

Based on the mixed reality test method in this paper, the ACC test steps of the self-driving vehicle are as follows:

Step 1:

- First, establish the mapping relationship between the virtual environment and the actual environment.
- Set up a road network map and corresponding facilities in the simulation environment.
- Establish the communication connection between the vehicle model and the actual vehicle in the virtual environment.

Step 2: Assign tasks to the virtual background vehicle, including driving path, task type and attributes ID, the tested vehicle, and transmit to the actual background vehicle simultaneously.

Step 3: Start the test, the main vehicle accelerates from static to the target speed, and the background vehicle enters the running state according to the predetermined track and speed. The main vehicle's running data and the background vehicle's running data are collected in real-time, and the virtual vehicle is reflected in the simulation environment synchronously.

Step 4: If the test requirements are met, we will evaluate the performance of the tested self-driving vehicle; if the test requirements are not met, the test return to step 2.

#### B. Test Results And Evaluation

Then only take the cut-in scenario as an example to illustrate the test evaluation. Let the background vehicle cuts into the main lane from the adjacent lane at a certain distance ahead of the main vehicle, and Table 1 shows the cutting parameters. After the main vehicle and the background vehicle reach a stable driving state, the background vehicle drives out of the tested lane and collects the main vehicle's position, speed, speed change rate, gear, and vehicle lighting state as the evaluation basis, so far a test is completed.

TABLE I. TEST PARAMETERS IN THE BACKGROUND VEHICLE CUT INTO SCENE

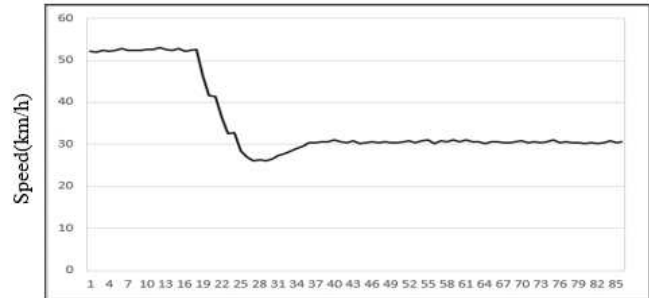
| Index | Main vehicle's speed(km/h) | Background vehicle's speed(km/h) | Time distance(s) | Cut in speed (km/h) |
|-------|----------------------------|----------------------------------|------------------|---------------------|
| 1     | 50                         | 30                               | 2~3              | 3.6~7.2             |
| 2     | 50                         | 60                               | 2~3              | 3.6~7.2             |
| 3     | 80                         | 60                               | 2~3              | 3.6~7.2             |

Take one test as an example, set the main vehicle speed as 50km/h, the background vehicle speed is 30km/h, the road condition is straight, and the background vehicle cuts into the main lane from the right front of the main vehicle. Record when the background vehicle starts the cut action as the cutting time, the main vehicle, and the background vehicle reach the stable state as the cutting completion time, from the cutting time to the cutting completion time as the cutting process. Table 2 is the operation data of the main and background vehicles before and after the cutting process. The data shows that both the main and background vehicles drive at a uniform speed before cutting in. After the cut, the main vehicle speed is down to 26.56km/h. During the cut, the main vehicle can complete automatic shift and braking. The minimum workshop distance is more than 2s, and the horizontal deviation of 6cm indicates that the vehicle body can be kept stable without noticeable shaking.

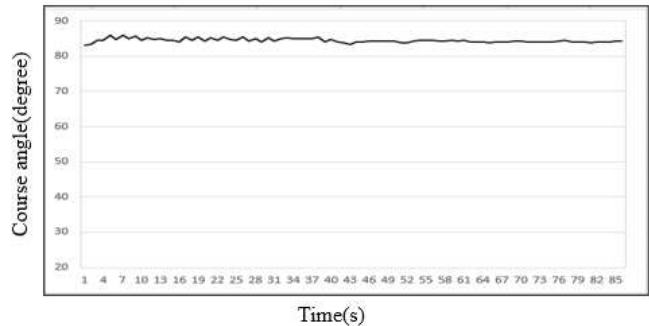
TABLE II. A SET OF TEST DATA BEFORE AND AFTER THE CUT IN ACTION

| Parameters                                             | Before the cut          |                         | After the cut           |                         |
|--------------------------------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|                                                        | Main vehicle            | Background vehicle      | Main vehicle            | Background vehicle      |
| Location coordinates (m)                               | (508972.73, 4115213.67) | (509006.01, 4115220.25) | (509135.96, 4115230.50) | (509166.34, 4115233.75) |
| Road radius (m)                                        | >500                    |                         |                         |                         |
| Course angle (degree)                                  | 83.3                    | 86.7                    | 84.1                    | 84.3                    |
| Speed (km/h)                                           | 52.12                   | 30.24                   | 26.56                   | 30.74                   |
| Acceleration ( $m/s^2$ )                               | 0.004                   | 0.002                   | -0.6                    | 0.01                    |
| Minimum workshop Time (s)                              | 2.3                     |                         | 2.1                     |                         |
| Maximum longitudinal centerline lateral deviation (cm) | -                       | -                       | 6                       | 314                     |
| Control mode                                           | Auto                    | Auto                    | Auto                    | Auto                    |
| out of auto or not                                     | Not                     | Not                     | Not                     | Not                     |
| Active braking or not                                  | -                       | -                       | Yes                     | -                       |
| switch the gears automatically or not                  | -                       | -                       | Yes                     | -                       |

As shown in Figure 4 (a) (b), at 18s, the main vehicle detected the background vehicle ahead that was cutting into the current lane, taking correct braking measures, and the vehicle speed slowed down significantly. Due to the straight road, the course angle of the main vehicle did not change significantly in the process. With the completion of the background vehicle cutting action, the main vehicle speed drops to a minimum value of 26 km/h at the 30s. The workshop distance gradually increases, the main vehicle automatically improves the speed to be close to the background vehicle speed, about 30km/h, and the two vehicles reach a stable operation state.



(a)The main vehicle's speed



(b)The main vehicle's course angle

Figure 4. The movement state of the main vehicle in the process of background vehicle cutting in

The analysis of the above test data shows that after the background vehicle cuts into the tested lane, the main vehicle can correctly recognize the cutting action of the front vehicle and automatically adjust the speed according to the speed of the front vehicle. After the main vehicle enters the stable state, it keeps a certain distance from the background vehicle in front. The stable speed is within the limited range, and the lateral deviation from the longitudinal centerline of the front vehicle is within the limited range. There is no excessive braking and acceleration phenomenon in the whole process. It is considered that the main vehicle has passed the cut in scene test one time. Repeat the test 20 ~ 30 times. If the passing rate exceeds 90%, it is considered that the main vehicle has passed the scene test.

## V. CONCLUSIONS

With the development of self-driving technology, the traditional single technology means such as virtual testing and road testing have been difficult to meet the test

requirements in complex application scene. This paper proposes a mixed reality-based self-driving test method and gives its system architecture and test flow. Finally, a specific case simply verifies the method, which provides a new idea for self-driving test technology. At present, scene-based self-driving testing technology is still in the exploration stage. It needs further research in many aspects, such as test mode, test scene library construction, test toolchain design, test case design, to provide strong support for developing the self-driving technology industry.

#### REFERENCES

- [1] Xiang-mo Zhao, Jing-jun Cheng, Zhi-gang Xu, et al. "An Indoor Rapid-Testing Platform for Autonomous Vehicle Based on Vehicle-in-the-loop Simulation". *China Journal of Highway and Transport*. 2019,32(6):124-134.
- [2] Bing Zhu, Pei-xing Zhang, Jian Zhao, et al. "Review of Scenario-Based Virtual Validation Methods for Automated Vehicles". *China Journal of Highway and Transport*. 2019,32(6):1-19.
- [3] E. Armagan and T. Kumbasar, "A fuzzy logic based autonomous vehicle control system design in the TORCS game environment", *Pamukkale University Journal of Engineering Sciences*, 2018,24(8):1435-1442.
- [4] LiLi, Wu-Ling Huang, Yue-hu Liu, et al. "Intelligence Testing for Autonomous Vehicles: A New Approach", *IEEE Transactions on Intelligent Vehicles*, 2016,1(2):158-166.
- [5] Wu-Ling Huang, Kun-feng Wang, LvYisheng, et al. "Autonomous Vehicles Testing Methods Review", 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC). New York: IEEE, 2016:163-168.
- [6] Yu Chen, Shitao Chen, Tangyike Zhang, et al. "Autonomous Vehicle Testing and Validation Platform: Integrated Simulation System with Hardware in the Loop", 2018 IEEE Intelligent Vehicles Symposium (IV) Changshu, Suzhou, China, June 26-30, 2018 :949-954.
- [7] O. Gietelink, J. Ploeg, B. De Schutter, et al. "Development of advanced driver assistance systems with vehicle hardware-in-the-loop simulations", *Vehicle System Dynamics*, 2006, 44(7):569-590.
- [8] Jiaqi Ma, Fang Zhou, Zhitong Huang, et al. "Hardware-in-the-loop Testing of Connected and Automated Vehicle Application: A Use Case for Queue-aware Signalized Intersection Approach and Departure", *Transportation Research Record*, 2018(2672):36-46.
- [9] Feilhauer Marius, Haering Juergen, Wyatt Sean. "Current Approaches in HiL-Based ADAS Testing", *SAE International Journal of Commercial Vehicles*, 2016:63-69.
- [10] Sukru Yaren Gelbal, Santhosh Tamilarasan, Mustafa Ridvan Cantas, et al. "A Connected and Autonomous Vehicle Hardware-in-the-loop Simulator for Developing Automated Driving Algorithms", *Proceedings of the 2017 IEEE International Conference on Systems, Man, and Cybernetics*. New York: IEEE, 2017:3397-3402.
- [11] H. Zhang, Y. Zhang, and C. Yin, "Hardware-in-the-loop simulation of robust mode transition control for a series-parallel hybrid electric vehicle", *IEEE Transactions on Vehicular Technology*, 2016,65(3):1059-1069.
- [12] Caroline Bianca Santos Tancredi Molina, J.R Almeida, Lucio F.Vismari, et al. "Assuring Fully Autonomous Vehicles Safety by Design: The Autonomous Vehicle Control Module Strategy", *IEEE, 2017 47th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshop*, 2017:16-21.
- [13] X. Li, Z. Sun, D. Cao, et al. "Real-time trajectory planning for autonomous urban driving: framework, algorithms, and verifications", *IEEE/ASME Transactions on Mechatronics*, 2016,21(2):740-753.
- [14] LiLi. "Auto Smart Turing Test: What Tests Does an Unmanned Car Test "Driver's License" Pass". [2016-10-20]. <https://www.ednchina.com/news/2016102000007.html>.